

Sampat Choudhary

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Education

Indian Institute of Technology, Kanpur
Bachelors of Technology in Electrical Engineering

Kanpur, IN
2016 - 2020

Work Experience

Prodigal

Sr Software Engineer

Bangalore, IN

April 2025-current

- Built **ProAgent**, an enterprise AI voice agent builder enabling prompt engineers to design and deploy **production-ready conversational agents** for the U.S. debt collection industry.
- Engineered a **real-time voice streaming pipeline** integrating **LiveKit (WebRTC)**, **Deepgram STT**, and multi-provider **TTS (ElevenLabs, Cartesia, Google)** with intelligent failover, achieving **99.9% uptime**.
- Owned and built a **scalable FastAPI middleware service** enabling LLM tool-calling (CRM data retrieval, email/SMS messaging, and **secure payment processing**) to support real-time actions during live phone calls.
- Built a configuration management backend and internal **Agent Builder UI** to deploy and **version multi-agent systems**; integrated **Twilio and LiveKit APIs** to automate SIP trunk provisioning and phone number linking, reducing agent setup time from **hours to minutes**.

Microsoft

Software Engineer II

Bangalore, IN

Feb 2024 - April 2025

- Designed **user and S2S authentication and authorization layers** for the v2 Viva Goals services, and **implemented a generic authentication NuGet package** that provides APIs for secure user and S2S authentication across all Viva Goals services and beyond, using Microsoft's standard .NET libraries.
- Worked with various internal teams to enable authentication for the new Viva Goals application and its services, such as Copilot and other integrations.
- Implemented **Continuous Access Evaluation** for Viva Goals to enhance security and make the product ready for selling to the Financial Services Industry.
- Became the SME and owner for authentication for Viva Goals services. Learned about Microsoft's security standards and best practices for **user authentication, S2S authentication, PoP authentication, client credential grant, and Continuous Access Evaluation** (allowing admins to revoke user access at any time), etc.
- Built a background worker host service using **Azure Event Hub** and **Azure Functions** to asynchronously push audit logs from the Viva Goals application to the M365 audit service, implementing CI/CD pipelines to deploy the service to Azure Kubernetes.

Software Engineer I

Bangalore, IN

Nov 2021 - Feb 2024

- Designed and implemented a scalable and extensible **permission model** for the OKR application, managing resource permissions across users, groups, and teams. Became the **SME** for the app's permission model.
- Developed an **ASP.NET Core authentication** service using **MISE and SAL** NuGet packages to enhance security for non-.NET applications. Owned the full lifecycle, including **design, development, CI/CD pipelines, and AKS deployment**.
- Optimized application performance by identifying and redesigning slow-running APIs, background jobs, and complex database queries.
- Collaborated with **Microsoft AuthNZ** to enhance authentication security (**PFT tokens**) and build extensible S2S utilities; partnered with Copilot to implement Substrate LLM API auth, supporting GA readiness.

Ally.io

Software Engineer

Chennai, IN

Aug 2020-Oct 2021

- Built and enhanced end-to-end features in **Ruby on Rails, AngularJS, and jQuery** to revamp the admin app and improve Customer Support efficiency.
- Developed scalable backend solutions using **Sidekiq, Golang, and MongoDB**, gaining hands-on experience with background jobs and data-driven services.

Patent

- Conceived, designed, and implemented a patented solution (**Dynamic Control of Multi-Nested OKR Alignment, Patent No. MS 412647-US-NP**) by collaborating with cross-functional teams to identify patentable ideas and navigate the filing process, contributing a unique innovation to Microsoft.

Skills and Interests

- **Programming Languages:** Python, C#, Go, Ruby, JavaScript
- **Backend & APIs:** ASP.NET Core, FastAPI, Ruby on Rails, REST APIs
- **AI Systems:** AI Agents, Voice AI Systems, Deepgram (STT), ElevenLabs (TTS), LiteLLM, LiveKit (WebRTC)
- **Databases:** PostgreSQL, MongoDB, Redis, Azure Cosmos DB
- **Messaging & Streaming:** Apache Kafka, AWS SQS, Azure Event Hub
- **Cloud & Infrastructure:** Azure, AWS, Docker, Kubernetes, Azure Pipelines for CI/CD, GitHub Actions
- **Developer Tools:** Claude-Code, Cursor